

In OpenStack, **over-commitment** is a resource management strategy that allows cloud operators to maximize the efficiency of their hardware by allocating more logical resources to virtual machines than are physically available on the host 1.

Definition of Over-commitment

Over-commitment refers to the practice of **allocating more resources (specifically CPU and memory) than the physical hardware possesses** 1. This strategy relies on the principle of **statistical multiplexing**, which assumes that not all virtual machines (VMs) will utilize their maximum allocated resources simultaneously 1.

CPU Over-commitment

- **Virtual to Physical Ratio:** This involves assigning more virtual CPUs (vCPUs) to instances than there are physical CPU cores on the compute node 1.
- **Hyper-threading:** To support higher levels of CPU over-commitment, **hyper-threading** is highly recommended 2. This technology effectively doubles the number of existing cores recognized by the system, allowing for a higher density of VM placement without immediate physical exhaustion 2.

Memory Over-commitment

- **RAM Allocation:** Similar to CPU, memory over-commitment allows the sum of the RAM allocated to all running instances on a host to exceed the actual physical RAM installed on that host 1.
- **Sizing Importance:** Because memory is a finite resource that is harder to "swap" than CPU cycles without extreme performance hits, the sources emphasize that **sizing compute nodes correctly** is critical to avoid waste or system instability 3.

Advantages

- **Improved Resource Utilization:** It prevents "waste" by ensuring that powerful physical nodes are not left idle when VMs are only using a fraction of their assigned capacity 1, 3.
- **Cost Efficiency:** By hosting more instances on fewer physical servers, organizations can reduce the total amount of hardware required for their cloud infrastructure 2.

Risks and Performance Tradeoffs

- **Simultaneous Utilization:** The primary risk is **performance degradation** 1. If multiple VMs on an overcommitted host heavily utilize their resources at the same time, the physical hardware will be unable to meet the demand, leading to slow processing and latency 1.
- **Capacity vs. Performance:** There is a constant tradeoff between **capacity** (hosting more VMs) and **performance** (ensuring high speed and reliability for each VM) 4.
- **Resource Exhaustion:** If statistical multiplexing fails (i.e., all users peak at once), the system may face total resource exhaustion 1.

Exam-Focused Explanation

1. **Key Concept:** Over-commitment is built on **statistical multiplexing**—the bet that not everyone needs 100% of their resources at the same time 1.
2. **Recommended Hardware:** Always mention **hyper-threading** as a way to double available cores for over-commitment purposes 2.
3. **The "Catch-all" Problem:** There is no single formula for over-commitment that works for every cloud 3. Operators must:
4. **Estimate** sample calculations for CPU and RAM 3.
5. Consider **workload patterns** and organization **risk tolerance** before setting ratios 1.
6. **Placement Logic:** The **nova-scheduler** plays a role here by checking compute node availability to ensure that even in an overcommitted environment, a node has enough "free" logical space to accommodate a new VM without degrading existing ones 5.